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Clustering and Interpretation of Extreme Rainfall Events Using Multimodal Large Language Models and Retrieval-Augmented Generation: Based on Autumn Data from Northeastern Taiwan †

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Abstract

Extreme autumn rainfall has become frequent due to climate change, making disaster prevention increasingly difficult. We combined a retrieval-augmented generation (RAG) framework with a multimodal large language model (multimodal LLM) to automatically cluster and explain weather patterns. The multimodal LLM assists in selecting an appropriate clustering method, such as hierarchical clustering, to determine the optimal number of clusters. To enhance weather map interpretation and reduce hallucinations or uncertainty, 13 specialized prompt roles are designed to guide the model's reasoning process. The method is applied to autumn-season data from 1960 to 2019, using weather records from the Taiwan Climate Change Projection and Information Platform and the ERA5 reanalysis dataset by the European Center for Medium-Range Weather Forecasts. The results show that three dominant weather types were identified. The identified types are typhoon with companion system (TC_NE, 51%), northeasterly pattern (NE, 30%), and tropical cyclone (TC, 19%). The developed method in this study provides a new approach for interpreting extreme weather events under changing climate conditions.

Keywords: extreme rainfall; hierarchical clustering; large language model (LLM); retrieval-augmented generation (RAG); ERA5 reanalysis

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1. Introduction

Climate change has intensified in recent decades, resulting in a significant increase in both the frequency and severity of extreme weather events, which pose major challenges to disaster preparedness and weather forecasting systems [1]. Climate-related datasets are inherently complex, characterized by high dimensionality, non-linearity, and non-stationarity. The machine learning methods—such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs)—have shown strong performance in extracting spatial and temporal patterns [2], particularly in weather pattern classification and short-term rainfall prediction. However, these models still face limitations in multimodal data integration, knowledge reasoning, and interpretation of complex weather phenomena.

Under the influence of global warming, Taiwan has experienced a consistent rise in overall temperatures, with autumn temperatures notably higher. Additionally, typhoon activity has become more frequent during this season. As a result, Taiwan's autumn weather is often influenced by tropical cyclones or compound events, leading to increasingly extreme

and variable rainfall patterns. Studies have indicated that the frequency of extreme rainfall events in Taiwan has been rising, especially during winter, spring, and the typhoon season, with average extreme precipitation levels showing a positive trend. Notably, extreme rainfall during the typhoon season is highly correlated with the increasing frequency and duration of typhoons [3]. Since Taiwan's typhoon season often extends into autumn, the seasonal weather becomes unstable and susceptible to interactions among strong convection, low-pressure disturbances, and the northeast monsoon. These interactions increase the likelihood of severe rainfall or compound events [4]. Making autumn a high-risk period for disaster-level precipitation, thereby complicating weather forecasting and disaster preparedness.

Meteorological data typically exhibit strong spatiotemporal correlations and are presented in multiple formats, such as numerical values, visual maps, and textual descriptions, forming a classic multimodal data structure. Traditional statistical or machine learning methods often struggle to fully interpret both the spatial image content and the semantic meaning of meteorological patterns, leading to reduced accuracy in weather type recognition and classification. To address these challenges, we employed a multimodal large language model (LLM) with robust capabilities in processing complex data structures [5] to analyze reanalysis meteorological datasets. The model is queried to recommend appropriate clustering methods and determine the optimal number of clusters for identifying the diverse weather patterns observed in Taiwan during autumn. Furthermore, the study incorporates a retrieval-augmented generation (RAG) framework [6], which leverages the advantages of natural language processing (NLP) to reduce hallucination in generated content, thereby enhancing the accuracy and reliability of weather pattern analysis and interpretation.

Based on the model's recommendation, we applied hierarchical clustering [7] to cluster autumn meteorological conditions. Subsequently, each cluster's average weather pattern is analyzed and interpreted. This approach not only improves clustering accuracy but also contributes to a deeper understanding of Taiwan's autumn climate variability and the features of extreme events. Ultimately, it provides a scientific basis for future disaster mitigation and decision-making.

2. Data and Method

2.1. Data and Definition of Extreme Rainfall Events

We defined extreme rainfall events as those with daily precipitation exceeding the 95th percentile threshold. The analysis was conducted on the data from 1979 to 2020, focusing on the northeastern region of Taiwan (Keelung, New Taipei City, Taipei City, and Yilan County) [8].

2.2. Data Processing

A total of 187 days associated with extreme rainfall events were identified for analysis. Subsequently, ERA5 reanalysis data were employed to extract gridded atmospheric variables at a standardized spatial resolution of 0.5° [9,10]. To ensure consistency across datasets, linear interpolation was performed to resample the original data onto a uniform 0.5° grid [11].

On the temporal scale, four standard time points were selected for each day—00, 06, 12, and 18 coordinated universal time (UTC)—with data sampled at six-hour intervals. The four snapshots were then averaged to generate a daily mean weather map, representing the overall atmospheric condition for each selected day.

2.3. LLM

In this study, the selected large language model is the Gemma3-12B, a member of the Gemma3 family with approximately 12 billion parameters [12]. Compared with other LLMs that typically exceed tens or even hundreds of billions of parameters, Gemma3-12B

is relatively lightweight. It can be executed on a single GPU or TPU and supports edge deployment, which not only reduces computational costs but also enhances data security in practical applications.

The Gemma3 series is designed for multimodal processing, capable of handling both image and text inputs. This feature makes it particularly suitable for this study, which requires integration of weather map analysis and natural language generation. In the implementation workflow, prompts were carefully crafted to guide the LLM in recommending an appropriate clustering method and determining the optimal number of clusters based on the ERA5 reanalysis data [13]. Subsequently, the RAG system, built upon expert knowledge and historical data, was introduced to support the LLM in retrieving domain-specific information. This enabled the model to accurately identify and describe weather types associated with the 187 extreme rainfall events. The overall process is illustrated in Figure 1.

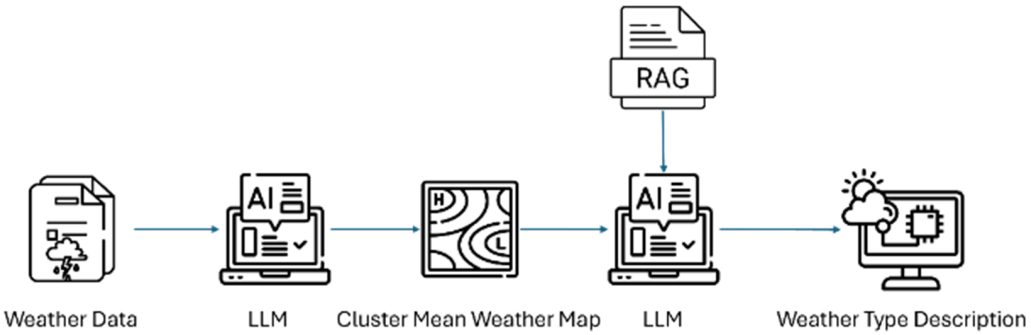


Figure 1. The workflow illustrates the proposed approach, starting from ERA5 meteorological data. A large language model (LLM) is used to recommend a suitable clustering method and the number of clusters, followed by the generation of representative weather maps. A Retrieval-Augmented Generation (RAG) mechanism is then integrated to assist in the interpretation process, ultimately producing explainable descriptions of weather patterns.

2.4. RAG

A custom-designed RAG knowledge base was developed, focusing on the distinguishing features and classification criteria of three major weather types: the northeast monsoon (NE), typhoon-related compound events (TC-NE), and typhoons (TC). When the model performs weather map recognition, it first retrieves relevant meteorological features from the knowledge base to guide its reasoning and classification process. This integration helps enhance interpretability and domain consistency during model inference. An example of the knowledge base rules is shown in Table 1. This knowledge base design enhances the model's reasoning accuracy in identifying weather patterns, reduces the risk of hallucination [14], and further supports its application in the analysis of extreme weather events.

Table 1. Example of internal rule design in the rag knowledge base (constructed based on expert knowledge).

Weather Type	Rule1	Rule2
TC	Presence of a clear cyclonic wind field (counterclockwise rotation)	Closed low-pressure center detected
NE	RH shows a north-dry/south-moist pattern (low RH in the north, slightly higher RH in the south)	No cyclonic wind field observed
TC-NE	Presence of northeast monsoon	Closed isobars or cyclonic wind field in the south indicates tropical disturbance or low-pressure trough
Anti-misclassification rules	High RH ≠ necessarily TC-NE	Absence of a closed low-pressure center ≠ not TC-NE

2.5. Role Prompt and System Prompt for LLM

In LLM applications, the role prompt is used to explicitly define the identity or role the model should assume [15], while the system prompt provides detailed decision-making rules to minimize the likelihood of incorrect judgments. We optimized the identification of compound events (TC-NE), as such cases often exhibit features of both typhoons (TC) and northeast monsoons (NE), making them prone to misclassification. The model may confuse compound types with either typhoon or monsoon types due to overlapping characteristics [16]. To address this challenge, the system prompt was carefully crafted to emphasize the unique meteorological features of compound events. Out of 13 designed instructions, five representative examples were selected and shown in Table 2 to assist the model in making more accurate judgments.

Table 2. Representative system prompt rules for identifying compound events (TC-NE) these rules were incorporated into the system prompt to assist the LLM in distinguishing compound weather patterns from typhoon or monsoon types.

Role	Content
1	If the wind field is predominantly easterly or northeasterly without a clear counterclockwise rotation or cyclonic center, it should not be classified as a cyclonic wind field.
2	To determine convergence, the presence of wind direction convergence and a distinct intersection center must be observed. Slightly curved wind flow alone does not constitute convergence.
3	If isobars are only locally curved or form a semi-circular shape without a fully closed low-pressure center, it should not be considered a closed low. The assessment must confirm whether continuous isobaric lines form a closed loop around a center.
4	If the wind field is dominated by northeasterly winds, with relative humidity (RH) being dry in the north and moist in the south, and the mean sea level pressure (MSLP) shows a gradient from high in the east to low in the west without a closed center or convergence, it should be classified as NE.
5	When assessing convergence, the presence of a clear wind direction convergence and an identifiable intersection center must be observed. Slightly curved wind flow does not meet the criteria for convergence.

In this study, the role prompt for LLM was defined as: "You are a professional meteorologist." This was combined with the system prompt designed earlier to provide explicit decision-making criteria. Such a prompt strategy enhances the model's ability to accurately classify weather patterns, particularly in complex extreme weather scenarios, and significantly reduces the risk of misclassification while improving the consistency and reliability of the model's interpretations [17].

3. Result and Discussion

We used the ERA5 reanalysis dataset, which provides meteorological variables including air temperature, wind components, sea level pressure (SLP), and relative humidity (RH) at 6 h intervals. To ensure spatial consistency with subsequent processing, the original resolution ($0.25^\circ \times 0.25^\circ$) was resampled to a coarser $0.5^\circ \times 0.5^\circ$ grid using linear interpolation techniques [18]. After spatial adjustment, four standard synoptic times (00, 06, 12, and 18 UTC) were averaged for each day to obtain the daily mean synoptic weather patterns, which serve as the basis for clustering and subsequent classification.

The detailed experimental procedures and results are described in the following subsections.

3.1. Input Variable Processing and Model Recommendation

Five meteorological variables were selected as input features, including air temperature, relative humidity, zonal wind (U), meridional wind (V), and mean sea level pressure (MSLP). These variables jointly represent both the thermodynamic and dynamic structures of synoptic weather patterns. The selected variables were fed into LLM, which was then

prompted to recommend a suitable clustering method for classifying distinct weather types. Based on the model's analysis, hierarchical clustering was identified as the most appropriate technique for this study [7].

This method is particularly well-suited for analyzing high-dimensional meteorological data, as it effectively captures the hierarchical relationships among samples. By computing pairwise distances (e.g., Euclidean distance) and iteratively merging clusters, hierarchical clustering allows the construction of a dendrogram that reflects the underlying structure of the data. Such a technique is ideal for uncovering spatial or temporal variations in weather regimes and for exploring potentially latent classification structures within complex climate datasets.

3.2. Clustering Implementation and Determination of Optimal Cluster Number

Following the model's recommendation, hierarchical clustering was used to group the daily weather data. A dendrogram was constructed to visualize the similarity and hierarchical structure among samples. As shown in Figure 2, the dendrogram reveals how different samples are merged across various distance thresholds, reflecting the nested relationships within the data. To determine the optimal number of clusters, the dendrogram image was fed back into LLM to assess the most suitable cluster count and explain the rationale. The model responded, "The optimal number of clusters is 3. This is because at a distance threshold of approximately 15, three distinct branches emerge, indicating that a three-cluster division provides representative groupings" (Table 3).

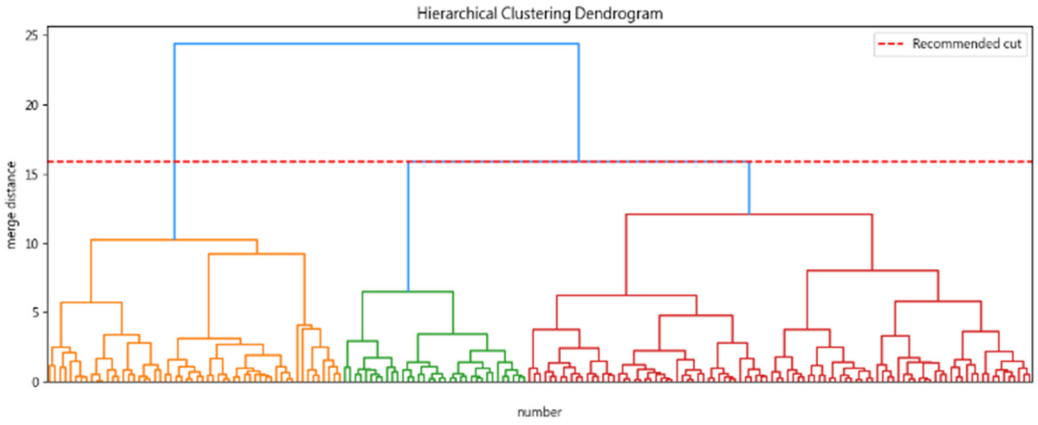


Figure 2. Hierarchical clustering dendrogram. Branch colors indicate the three optimal clusters (k = 3) obtained by cutting the dendrogram at the recommended linkage distance (red dashed line). The clusters are interpreted as: red—Cluster 1 (TC-NE), orange—Cluster2(NE), and green—Cluster 3 (TC).

Table 3. LLM recommendation on optimal number of clusters and rationale.

Prompt	LLM Response
Please review this dendrogram and determine the optimal number of clusters. Provide a single number along with your reasoning.	From the dendrogram, three distinct branches are clearly observed at a distance threshold of approximately 15. This suggests that dividing the data into three groups may be a more suitable choice.

3.3. Cluster Analysis and Weather Pattern Identification

After the LLM recommended dividing the data into three clusters, the mean weather map for each group was generated. These composite maps were then fed back into the LLM, combined with the RAG framework and specifically designed prompts, to facilitate automatic identification and interpretation of weather patterns.

Based on the synthesized meteorological features—including wind field structures (UV), spatial distribution of RH, and MSLP—the three clusters were identified as follows:

- Cluster 1: TC-NE (Figure 3)

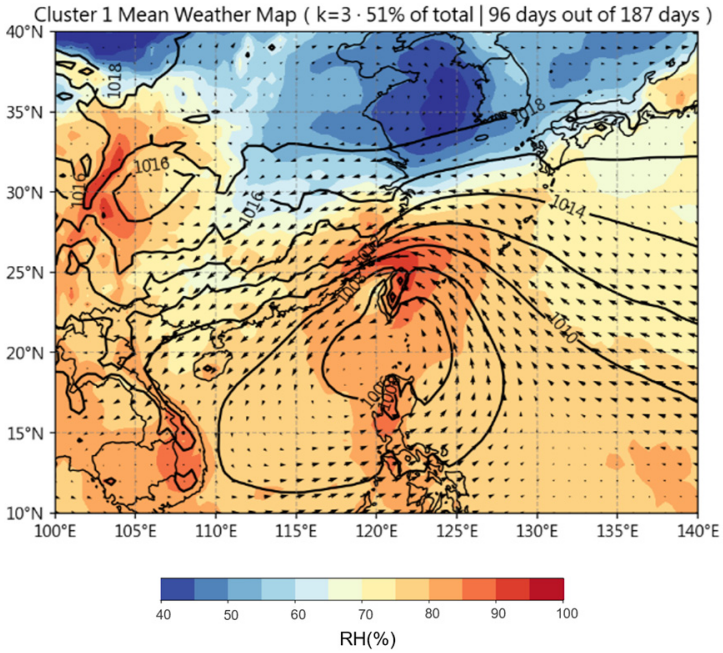


Figure 3. Mean synoptic patterns and the relative frequency of three clustered autumn weather types in northeastern Taiwan. Cluster 1 (Accompanied type; TC–NE) accounts for 51% of cases. Shading indicates relative humidity (RH, %), black labeled contours denote sea level pressure (SLP, hPa), and wind vectors show the 850-hPa horizontal wind field, with vector orientation indicating wind direction and vector length indicating wind speed.

- This cluster represents the most frequently occurring synoptic pattern in the classification, accounting for approximately 51% of the total samples. LLM identified this type based on several key features: a high-humidity zone (RH > 80%) extending from the south toward northern Taiwan, forming a distinct gradient with drier air over the northern region—indicative of a south-wet/north-dry configuration. At the 850 hPa level, the wind field is predominantly northeasterly, but significant wind convergence and deformation are observed near southern Taiwan, along with mild curvature in the isobars. While no clearly defined closed low-pressure system is present, the overall synoptic structure reveals an interaction between the northeasterly monsoon and a tropical disturbance. This aligns with the typical characteristics of a Typhoon–Northeasterly Monsoon Interaction (TC-NE) pattern.
- Cluster 2: NE (Figure 4)
- This cluster accounts for approximately 30% of the classified samples and represents a typical northeast monsoon (NE) synoptic pattern. At the 850 hPa level, wind direction is predominantly northeasterly, while the relative humidity (RH) exhibits a north-dry/south-wet distribution: RH values range from 70 to 100% in the south and decrease to 40to 60% in the north. The mean sea level pressure (MSLP) field displays an east-high/west-low gradient without any closed low-pressure systems. Although a moist zone is slightly advancing northward from the south, there is no apparent wind convergence or tropical disturbance present. These features are consistent with the classic characteristics of a northeast monsoon (NE) synoptic condition.

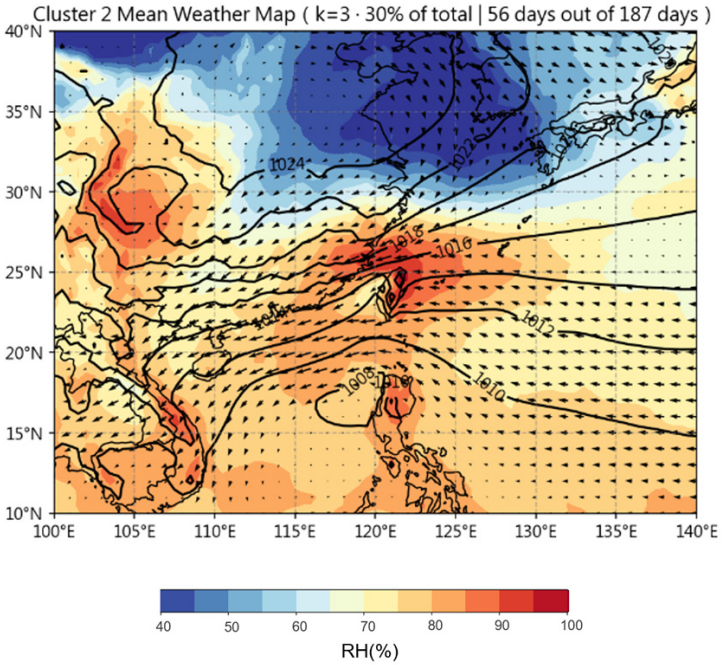


Figure 4. Mean synoptic patterns and the relative frequency of three clustered autumn weather types in northeastern Taiwan. Cluster 2 (Northeast monsoon type; NE) accounts for 30% of cases. Shading indicates relative humidity (RH, %), black labeled contours denote sea level pressure (SLP, hPa), and wind vectors show the 850-hPa horizontal wind field, with vector orientation indicating wind direction and vector length indicating wind speed.

- Cluster 3: Typhoon (Figure 5)

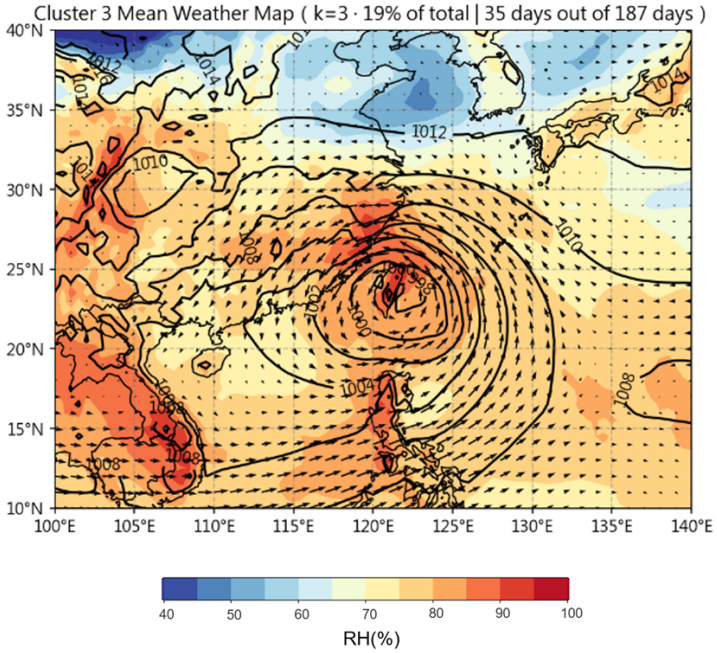


Figure 5. Mean synoptic patterns and the relative frequency of three clustered autumn weather types in northeastern Taiwan. Cluster 3 (Typhoon type; TC) accounts for 19% of cases. Shading indicates relative humidity (RH, %), black labeled contours denote sea level pressure (SLP, hPa), and wind vectors show the 850-hPa horizontal wind field, with vector orientation indicating wind direction and vector length indicating wind speed.

- This cluster accounts for approximately 19% of all samples—the smallest among the three groups—but exhibits distinct characteristics of a tropical cyclone. The relative humidity (RH) field shows high moisture concentrations over southern and eastern offshore regions. The wind field presents a clear counterclockwise cyclonic circulation pattern, with a central sea-level pressure of approximately 1002 hPa enclosed by concentric isobars, indicating a well-defined closed low-pressure system. No northeast monsoon signal is detected, nor is there evidence of interaction with monsoonal flows. Therefore, this group is classified as a typical typhoon-type (TC) synoptic condition.

Based on the model's interpretation of the three clusters, it is evident that the autumn weather patterns in Taiwan are significantly influenced by the interaction between the northeast monsoon and tropical disturbances. As illustrated in Figure 6, among the identified types, the accompanied type (TC-NE) is the most frequently observed, suggesting that the coexistence of typhoons and northeast monsoonal flows is a common synoptic feature during the autumn season.

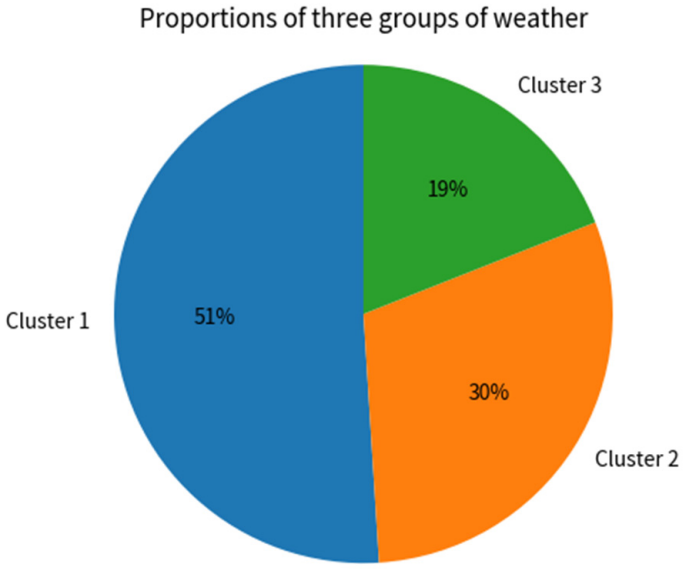


Figure 6. Pie chart showing the proportion of each cluster within the total number of samples. Blue represents Cluster 1 (TC-NE), orange represents Cluster 2 (NE), and green represents Cluster 3 (TC).

4. Conclusions

We developed a two-stage framework that first determines the optimal number of weather clusters using LLM, followed by the integration of multimodal LLM with RAG architecture to identify three dominant autumn weather patterns in northeastern Taiwan: TC, TC-NE, and NE. Leveraging a customized meteorological knowledge base, the framework enables effective interpretation of synoptic features—such as wind fields, relative humidity, and sea level pressure—facilitating accurate weather-type classification and semantic understanding. The results highlight the potential of LLM-based frameworks in professional meteorological applications, particularly in recognizing complex synoptic patterns from weather maps.

This method addresses the limitations of traditional clustering approaches, which often struggle to distinguish ambiguous or overlapping weather types. Notably, it enhances the model's ability to identify accompanied events (TC-NE), which involve interactions between tropical cyclones and northeast monsoons—phenomena commonly observed during autumn in Taiwan. As autumn is characterized by significant synoptic variability, the proposed system helps reveal the underlying structures that contribute to

extreme rainfall events, providing valuable insights for disaster risk reduction and climate impact assessment.

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